Assignment #3: Designing a Bayesian Network for a Pakistani Real World Problem

Huzaifa Iqbal (31563)

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1 Problem Description

Topic: Urban Commute Reliability and Student Attendance at IBA Karachi

This Bayesian Network models the uncertainty in a student's daily commute the university and the resulting impact on their attendance. Situated purely in the Pakistani and IBA context, this model addresses the systemic inefficiencies of Karachi's infrastructure. It explores how a universal scheduling problem, ensuring students arrive on time for lectures, is disrupted by multiple external factors such as sudden political protests, severe monsoon weather, and ongoing red and green Line construction. The network evaluates how some strategies (like leaving early or choosing a bike) affect these larger systemic disruptions.

2 Variables and Justifications

The network consists of 10 key variables, divided into environmental factors, traffic conditions, personal choices, and final outcomes.

Variable	States	Description	Justification
1. Rain (R)	True, False	Occurrence of monsoon/unexpected rain.	Heavily impacts Karachi's road infrastructure and drainage.
2. Protest (P)	True, False	Sudden political or social unrest.	Causes sudden road closures and diverts traffic unpredictably.
3. Peak Hours (H)	True, False	Commuting during 8:00 AM or 5:00 PM.	This is when students are coming to university or going home from university and this is when the traffic is the most.
4. Construction (C)	True, False	Ongoing BRT (Red/Green Line) construction work.	causes a lot of traffic.
5. Traffic Jam (T)	High, Low	Overall congestion on the route.	Rain, Protests, Peak Hours and Construction all add to traffic jams.
6. Accident (A)	True, False	Occurrence of a vehicle breakdown or collision.	Could cause major or minor delays depending on severity of accident.
7. Transport Mode (M)	Car, Bike, Bus	The student's chosen method of travel.	Different modes (e.g., Bykea vs. Shuttle) means different time taken to travel.
8. Early Departure (E)	True, False	Leaving home with some extra time in hand.	A personal control factor that affects arriving late or on time.
9. Arrival Status (S)	OnTime, Late	Arrival of the student at campus.	The primary goal and physical outcome of the entire commute process.
10. Marked Absent (MA)	True, False	The final output of all the previous variables, is the student on time or are they late.	Depends on arrival status and the specific instructor's attendance policy.

3 Structure Design (Causal Relationships)

The Directed Acyclic Graph (DAG) defines the cause-and-effect flow of the commute.

- **R → T, P → T, H → T, C → T:** Rain, Protests, Peak Hours, and Construction are independent parent nodes that collectively cause Traffic Jams.
- **T → A:** High traffic increases the probability of minor collisions or overheating vehicles breaking down (Accidents).
- **T → S, A → S, M → S, E → S:** Traffic levels, the presence of an Accident, the chosen Transport Mode, and Early Departure directly and simultaneously dictate the student's final Arrival Status. Early departure does not reduce traffic; rather, it provides a buffer against it at the point of arrival.
- **S → MA:** Arriving late is the direct cause of being marked absent.

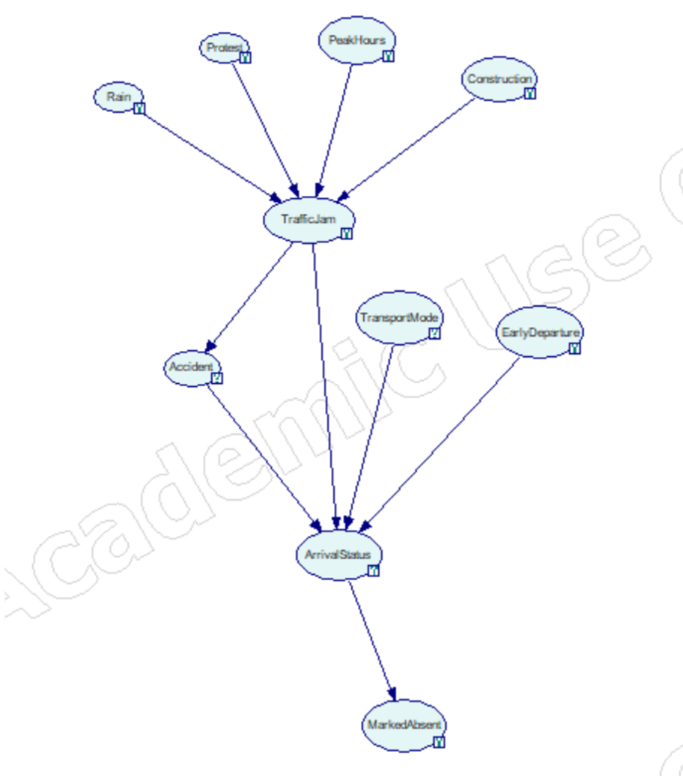

Figure 1: Directed Acyclic Graph (DAG)

4 Probability Specification

Conditional Probability Tables (CPTs) were populated using reasoned estimates of Karachi's urban dynamics.

Prior Probabilities (Independent Variables)

- **Rain (R):** True (0.05), False (0.95) — Rain is statistically infrequent.
- **Protest (P):** True (0.10), False (0.90) — Political instability means sudden protests but protests have also become rare.
- **Peak Hours (H):** True (0.50), False (0.50) — Split evenly to represent a standardized schedule. Students usually come in the mornings which is a peak hour and leave in the evening which is another peak hour.
- **Construction (C):** True (0.80), False (0.20) — Red/Green Line work is a near constant reality.
- **Early Departure (E):** True (0.70), False (0.30) — Assumes a proactive student attempting to optimize their schedule.
- **Transport Mode (M):** Car (0.40), Bike (0.40), Bus (0.20), many students either drive or carpool, or they use bykea. comparatively, less students travel by bus

Conditional Probabilities (Key Interactions)

- **Traffic Jam (T):** If Rain or Protests occur, $P(T = High) \geq 0.95$, regardless of time or construction, reflecting total gridlock. During normal Peak Hours with Construction, $P(T = High) = 0.85$.
- **Accident (A):** $P(A = True \mid T = High) = 0.15$; rises to 0.35 if traffic is low since cars speed when traffic is low.
- **Arrival Status (S):**
  - If an accident occurs ($A = True$), $P(S = Late) \geq 0.80$ across almost all modes and departure times, indicating system failure.
  - If $T = High, E = True, M = Bike$, $P(S = OnTime) = 0.85$, showing the resilience of two-wheelers in congestion.
  - If $T = High, E = False, M = Car$, $P(S = Late) = 0.85$.

Complete Probability Tables

Table 1: Prior: Rain (R)

State	**True**	**False**
Prob.	0.05	0.95

Table 2: Prior: Protest (P)

State	**True**	**False**
Prob.	0.10	0.90

Table 3: Prior: Peak Hours (H)

State	True	False
Prob.	0.50	0.50

Table 4: Prior: Construction (C)

State	True	False
Prob.	0.80	0.20

Table 5: Prior: Early Departure (E)

State	True	False
Prob.	0.70	0.30

Table 6: Prior: Transport Mode (M)

State	Car	Bike	Bus
Prob.	0.40	0.40	0.20

Table 7: CPT for Traffic Jam (T)

Rain	Protest	Peak Hours	Construction	High	Low
True	True	True	True	0.99	0.01
True	True	True	False	0.98	0.02
True	True	False	True	0.97	0.03
True	True	False	False	0.95	0.05
True	False	True	True	0.97	0.03
True	False	True	False	0.93	0.07
True	False	False	True	0.92	0.08
True	False	False	False	0.9	0.10
False	True	True	True	0.87	0.13
False	True	True	False	0.83	0.17
False	True	False	True	0.77	0.23
False	True	False	False	0.7	0.3
False	False	True	True	0.78	0.22
False	False	True	False	0.65	0.35
False	False	False	True	0.68	0.32
False	False	False	False	0.10	0.90

Table 8: CPT for Accident (A)

Traffic Jam	True	False
High	0.15	0.85
Low	0.35	0.65

Table 9: CPT for Arrival Status (S)

Accident	**Traffic Jam**	**Transport Mode**	**Early Dep.**	**OnTime**	**Late**
True	High	Car	False	0.15	0.85
True	High	Car	True	0.25	0.75
True	High	Bike	False	0.05	0.95
True	High	Bike	True	0.1	0.9
True	High	Bus	False	0.1	0.9
True	High	Bus	True	0.20	0.80
True	Low	Car	False	0.25	0.75
True	Low	Car	True	0.37	0.63
True	Low	Bike	False	0.15	0.85
True	Low	Bike	True	0.25	0.75
True	Low	Bus	False	0.15	0.85
True	Low	Bus	True	0.3	0.7
False	High	Car	False	0.45	0.55
False	High	Car	True	0.75	0.25
False	High	Bike	False	0.70	0.30
False	High	Bike	True	0.85	0.15
False	High	Bus	False	0.4	0.6
False	High	Bus	True	0.7	0.3
False	Low	Car	False	0.8	0.2
False	Low	Car	True	0.95	0.05
False	Low	Bike	False	0.85	0.15
False	Low	Bike	True	0.99	0.01
False	Low	Bus	False	0.75	0.25
False	Low	Bus	True	0.95	0.05

Table 10: CPT for Marked Absent (MA)

Arrival Status	**True**	**False**
Late	0.80	0.20
OnTime	1	0

5 Inference and Use Cases

The network was queried using GeNIe Modeler to observe posterior probabilities under specific evidence.

Use Case 1

- **Evidence:** `Rain = True, Protest = False, Construction = True, PeakHours = True, TransportMode = Car, EarlyDeparture = True`.

- **Interest:** Observing how a normal morning time commute with rain looks like when coming from university road
- **Results:** Setting Rain to True spikes the probability of `Traffic Jam = High` to nearly 97% with chance of an accident being 15%. Consequently, even with an `Early Departure` baseline of 70%, the posterior probability of `Arrival Status = Late` is 32% , leading to a small chance of being `Marked Absent`. The probability of being marked late is 26% because teachers usually are considerate and mark attendance at the end of class.

Use Case 2

- **Evidence:** `Rain = False, Protest = False, Construction = True PeakHours = True TransportMode = Bike, EarlyDeparture = False`.
- **Interest:** Testing if we leave late in the morning but we travel on bike which is fast and we are travelling through university Road, then what happens.
- **Results:** Despite heavy traffic and failing to leave early, setting the mode to "Bike" maintains a survivable probability of arriving OnTime (approx. 60%) compared to setting the mode to "Car" (45% OnTime). This accurately reflects the reality of ride hailing bikes navigating through traffic on University Road.

Use Case 3

- **Evidence:** , `Rain = False, Protest = True, Construction= False, PeakHours = False, TransportMode = Bus, EarlyDeparture = False`.
- **Interest:** We are now trying to see that if we have left late from our house and we travel through bus and there is a protest going on, then what are the chances we might be late.
- **Results:** For this, we see a higher chance of being marked absent, 47% chance. This is because due to protest, we are bound to find traffic with the probability of traffic jam being 70%. Also travelling through bus takes time since it is a slow and big vehicle with multiple stops.